Solution

Let $A(-5, 1)$ and $B(3, 2)$ be the given points

Let $P(x_1, y_1)$ be any point on the locus.

Given that $\angle APB = 90^o$.

Triangle APB is a right angle triangle.

$$BA^2 = PA^2 + PB^2$$

$$(3+5)^2 + (2-1)^2 = (x_1 + 5)^2 + (y_1 - 1)^2 + (x_1 - 3)^2 + (y_1 - 2)^2$$

$$65 = x_1^2 + 10x_1 + 25 + y_1^2 - 2y_1 + 1 + x_1^2 - 6x_1 + 9 + y_1^2 - 4y_1 + 4$$

i.e., $2x_1^2 + 2y_1^2 + 4x_1 - 6y_1 + 39 - 65 = 0$

i.e., $x_1^2 + y_1^2 + 2x_1 - 3y_1 - 13 = 0$

The locus of $P(x_1, y_1)$ is

$$x^2 + y^2 + 2x - 3y - 13 = 0$$

Exercise 3.1

1. Find the locus of a point which is equidistant from (1, 3) and x axis.
2. A point moves so that it is always at a distance of 4 units from the point (3, –2)
3. If the distance of a point from the points (2, 1) and (1, 2) are in the ratio 2 :1, then find the locus of the point.
4. Find a point on x axis which is equidistant from the points (7, –6) and (3, 4).
5. If $A(-1, 1)$ and $B(2, 3)$ are two fixed points, then find the locus of a point P so that the area of triangle APB = 8 sq.units.

3.2 System of Straight Lines

3.2.1 Recall

In lower classes, we studied the basic concept of coordinate geometry, like distance formula, section - formula, area of triangle and slope of a straight lines etc.

We also studied various form of equations of lines in X std. Let us recall the equations of straight lines. Which will help us for better understanding the new concept and definitions of XI std co-ordinate geometry.

Various forms of straight lines:

(i) Slope-intercept form

Equation of straight line having slope m and y-intercept 'c' is $y = mx+c$

(ii) Point- slope form

Equation of Straight line passing through the given point $P(x_1, y_1)$ and having a slope m is

$$y - y_1 = m(x - x_1)$$

(iii) Two-Point form

Equation of a straight line joining the given points $A(x_1, y_1), B(x_2, y_2)$ is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

In determinant form, equation of straight line joining two given points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

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